

ufacturing through imported systems. Renesas has 38 per cent share in the worldwide automotive controller market.

How much does India contribute to Renesas' worldwide revenues from automotive segment?

India accounts for less than 5 per cent of our automotive sales. The major reason is India's dependency on imported systems. The use of microcontrollers in cars is also distinctly lower than in overseas markets. In volume segment, a lot of cars are without power windows and automated seat adjustment. Volkswagen Jetta, one of standard cars in foreign markets, uses 70 microcontrollers. But the same model when launched in India is probably using only half this number.

Volume will, however, grow in India. So we have to be present there to learn and grow and adapt to this market. As cars become much more sophisticated with higher-end models using much more electronics, price points change. A 1\$ increase in electronics in any equipment is corresponding to two-three times higher retail pricing. As a consumer, you are today much more willing to pay more for better features. If you look in India, at the highest end of the market, is there any room for local mobile phone players? The higher end has Nokia and Apple iPhones. As the market increases in sophistication, for the consumer it's no longer the price issue.

Why are Indian manufacturers lagging behind in the high-end segment?

As mentioned earlier, Indian manufacturers are dependent on imported systems. A good example is the appliances market. It's having a situation whereby if LG and Samsung account for half of the 3 million air-conditioner market in India, the other 50 per cent is shared among 15 manufacturers. Each of them will sell 2000-5000 air-conditioners per month. That means there is no room for being competitive. The easiest way is to import the controller board, motor or whatever.

How can they compete globally?

If the manufacturers are able to add value by adopting local solutions supporting local development, then they will be having a different ballgame in terms of being competitive and creating market for themselves. This is what we are promoting. That's the reason we have come up with India-developed solutions so that Indian manufacturers are able to compete with global counterparts.

What are your plans for India?

We will continue to expand our internal resources because in India the dimension of electronics is not just the local market. We believe India will be the R&D lab of the world. By this conviction, a lot of our customers have set up design centres here and we are obliged to support the global platform by enhancing our resources.

By strengthening sales, marketing and technical support

in India, Renesas Electronics expects to grow its India business by over 20 per cent annually.

Do you ever see having your own design centre in India?

It's a step thing. Indian top-tier infotech companies grew up from initially providing documentation services. They have then migrated to be able to support certification, later development and now even hardware. Renesas is doing the same.

As our multinational customers in India go through the same phases, we are correspondingly increasing our technical knowhow internally to support this. I believe in one year's time by supporting the R&D lab of the world, we will be able to do two things: influence much more the global platform cultivation and develop ASSPs specifically for the Indian market. Next year, we will introduce the first super-low-power, super-efficient performance microcontroller specifically developed for the Indian market. Ninety per cent of the input in terms of influencing the development of this microcontroller is from here.

What are your engagement programmes for engineers?

So far our engagement has been internal engagement for our known customers. But with changing times, our approach is also changing. We are willing to increase our engagement all over.

We intend to get well-connected to our first-time users through various programmes and support ecosystem. We are ready with our University programme and we intend to extend it countrywide. We will increase our social media interaction and Web usability to support first-time users. We will enhance our e-commerce possibilities. By that, we will reach out to more first-time users.

Looking forward, in the next three years, which of your target segments do you see growing the most?

We expect all our targeted segments—automotive, industrial and consumer—to grow at double-digits pace. If the segments are growing, what's much more important for us is to find new applications within sub-segments. Last year was a grinder. Then we entered the water purifier market, which is a unique Indian business. Now we are entering the dessert cooler business. It's very exciting for us that we are able to look and find new applications that use our solutions.

There is a lot of talk about LED lighting. What will come will be streetlighting. But what will be the possibility for home lighting—a different environment? Our mission as an MCU-based solutions company is to look for the uniqueness. Here we create a solution to support that.

In smart grids, we have the leadership. Anything from the PV inverters to solar cells to meters is all supported by an MCU-based solution. Today, we have some pure metering companies in the market trying to support smart grid but the reality is it's an MCU-based solution. Connectivity is the way towards supporting the smart grid. The complexity of a smart grid can only be resolved with a true MCU solution. ●